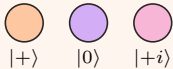
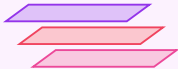


1. Prepare



product frame with $\bar{X}$ and the d^2 checks of S_0 deterministic

2. Measure



r rounds, all $2d^2 - 1$ checks, depth 6, one ancilla per check

3. Decode

S_0 matching

gauge detectors

correlated matching of both levels, after the last round

4. Recover

$$|\bar{+}\rangle$$
$$\bar{X} \mapsto (-1)^{\hat{\ell}} \bar{X}$$

Pauli frame update with the logical bit $\hat{\ell}$ and the gauge toggles

verification readout in the frame basis
reconstructs $\bar{X}$ and all of S_0